There are lots of indices flying around now, but the math isn't as hard as it looks. In fact, look at how the U matrices enter through the combination

$$\mathbf{U}^{\dagger}_{b''b}\ \mathbf{U}_{bb'}.$$

This combination is just the matrix product $U^{\dagger}U$. But recall that U is unitary. This tells you that the product $U^{\dagger}U$ is the unit matrix $\delta_{b''b'}$. As before, this amounts to an instruction to include all the terms where $b'' = b'$, and to ignore all the others. With this simplification, we get

$$\rho_{aa'} = \sum_{b}\ \psi^{*}(a'b)\ \psi(ab).$$

This is exactly the same as Eq. 7.31. In other words, $\rho_{aa'}$ *is exactly the same as it was before* U *acted*. Nothing that happens at Bob's end has any immediate effect on Alice's density matrix, even if Bob and Alice are maximally entangled. This means that Alice's view of her subsystem (her statistical model) remains exactly as it was. This remarkable result may seem surprising for a maximally entangled system, but it also guarantees that no faster-than-light signal has been sent.

7.10 THE QUANTUM SIM: AN INTRODUCTION TO BELL'S THEOREM

It's interesting that unitarity played a prominent role in guaranteeing that no signal can be sent instantaneously. If U had not been unitary, Alice's final density matrix *would indeed* have been affected by Bob.

What was it, then, that disturbed Einstein so much that he spoke of spooky action at a distance? To answer this question, it's important to understand that he and Bell were talking about a totally different notion of locality. To illustrate this, I am going to invent a computer game. What my new computer game does is try to fool you into thinking there is a quantum spin in a magnetic field inside the computer. You get to do experiments to test this possibility. See Fig. 7.1 for a schematic.